Cairo University

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FGSSR - Academic Study Platform

A Proposal Presented for Fulfillment

of The Diploma Project in Computer Science

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Introduction

In line with Egypt’s vision toward digital transformation (Egypt 2030) and the university’s commitment to academic quality and sustainable development, this project aims to create an integrated educational and academic environment within the faculty. The platform provides an effective study and communication tool, facilitating interaction among students, faculty, and administrative staff, enhancing educational efficiency, and supporting higher education digital transformation in line with global developments in educational technology.

Project background

Educational institutions often face challenges in providing fast and unified communication between students, faculty, and administrative staff. While some rely on emails, phone calls, or social media, these methods are often fragmented and inefficient.

The **FGSSR Academic Study Platform** introduces a centralized study and communication system, allowing students to access course-related materials, communicate with professors, and receive administrative support, improving information sharing, academic efficiency, and student engagement.

Problem statement

- Slow response to students’ academic inquiries.
- Difficulty in contacting professors outside class hours.
- Dependence on traditional methods such as printed announcements or individual phone calls.
- Absence of a unified digital academic channel connecting students, faculty, and administrative staff.

Goal

The **main goal** of this project is to provide an official, fast, and effective academic study and communication platform for students, faculty, and administrative staff, supporting academic quality and digital transformation.

Objectives

1. Provide a formal and fast communication channel for academic and administrative matters.
2. Reduce reliance on paper-based procedures and individual phone calls.
3. Support the faculty in achieving academic quality standards.
4. Contribute to implementing digital transformation in education in line with Egypt 2030.
5. Improve the efficiency of academic and administrative operations.
6. Facilitating access to course materials through material page.
7. Facilitating communication with the instructor via chat.
8. Facilitating access to external links for lectures.
9. Facilitating the instructor's ability to see the number of students registered in the course.
10. Facilitating the instructor's ability to see the course they will be teaching during the semester.
11. Facilitating access to exam schedules.
12. Facilitating access to lecture locations

Beneficiaries

The primary beneficiaries are **students, faculty members, and administrative staff**, who will gain access to a unified academic platform for studying and communication.

Proposed methodology.

1. Design the pages and user forms of the platform.
2. Create the central academic database.
3. Manage the platform structure.
4. Develop the frontend and backend code.
5. Populate and manage academic content.
6. Prepare the final package including platform pages, database files, and resources.

Languages and Libraries:

- PHP [1]
- MySQL [2]
- Laravel [3]
- JavaScript [4]
- jQuery [4]
- HTML5 [5]
- CSS [5]
- Bootstrap [6]
- Font Awesome [7]

Tools and Frameworks:

- GitHub for source control [8]
- Justinmind Prototyping Platform [9]
- TortoiseSVN [10]
- XAMPP [2]
- VS Code [8]
- Canva (for UX/UI design) [19]
- Stack Overflow (for debugging and reference) [11]
- MDN Web Docs (Mozilla) [12]
- W3C Standards [13]
- PHP The Right Way [14]
- Laracasts (tutorials) [15]
- CSS-Tricks [16]

- Node.js[18]

Project Plan

1. Gather requirements and analyze current academic communication issues.
2. Design the academic database.
3. Design the screens and dashboards for different user roles.
4. Build the database.
5. Implement frontend and backend code.
6. Test the platform functionality.
7. Enhance communication features.
8. Develop the File Management System module.
9. Develop the Messaging System module.
10. Implement the notification system.
11. Develop the user authentication and authorization module.
12. Build the course and content management module.
13. Implement the scheduling and timetable module.
14. Integrate analytics and reporting features.
15. Prepare documentation and user training materials.

References

[1] PHP Official Documentation – https://secure.php.net/

[2] MySQL Official Documentation – https://www.mysql.com/

[3] Laravel Framework – https://laravel.com/

[4] JavaScript & jQuery – https://jquery.com/

[5] HTML & CSS Tutorials – http://www.w3schools.com

[6] Bootstrap – http://getbootstrap.com/

[7] Font Awesome Icons – http://fontawesome.io/

[8] GitHub – https://github.com/

[9] Justinmind Prototyping Platform – https://www.justinmind.com/download

[10] TortoiseSVN Source Control – https://tortoisesvn.net/downloads.html

[11] Stack Overflow – https://stackoverflow.com/

[12] MDN Web Docs (Mozilla) – https://developer.mozilla.org/

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[16] CSS-Tricks – https://css-tricks.com/

[17] FreeCodeCamp – https://www.freecodecamp.org/ [18] GeeksforGeeks – https://www.geeksforgeeks.org/